# Separation Logic II

## Inductive predicates

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Lecture 11

COMSM0067 — Advanced Topics in Programming Languages
University of Bristol

# Where we are

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We cannot write

$$x \mapsto _ * _ \mapsto _ * _ \mapsto _ * \cdots$$

We need recursive heap assertions.

# Inductive predicates

# Linked lists, inductively

A cell $x \mapsto v, z$ holds a value $v$ and a next-pointer $z$:



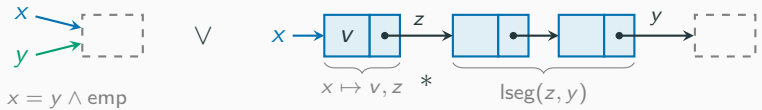
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**List segment from $x$ to $y$**

$$\mathsf{lseg}(x, y) \equiv (x = y \wedge \mathsf{emp}) \vee (x \neq y \wedge \exists v, z.\ x \mapsto v, z * \mathsf{lseg}(z, y))$$



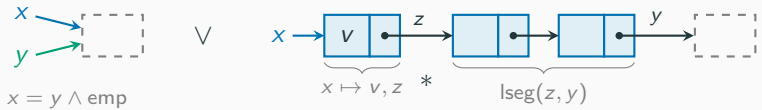
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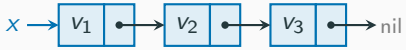
$$\mathsf{lseg}(x, y) \equiv (x = y \wedge \mathsf{emp}) \vee (x \neq y \wedge \exists v, z.\ x \mapsto v, z * \mathsf{lseg}(z, y))$$




Note the $*$ in the recursive case: the head cell is disjoint from the rest of the list. Separation is baked into the datatype — that is what makes it a *well-formed* list and not a tangle of aliases.

# Question: a full list

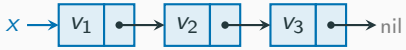
A full list is null-terminated: follow the next-pointers from $x$ and you reach nil.




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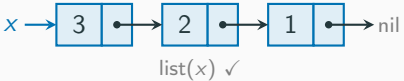
$$\mathsf{list}(x) \;\equiv\; \mathsf{lseg}(x, \mathsf{nil})$$

A full list is a segment that ends at nil. Including the empty list: $x = \mathsf{nil}$, no cells.

# list forgets the contents

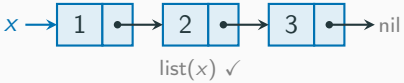
Both heaps satisfy list($x$):

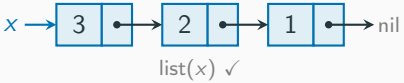


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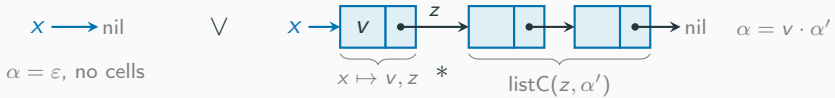

Enough for memory safety. Not enough to say *which* values the list holds.

# Tracking the *contents*

Index the predicate by the sequence of values $\alpha$ it holds:

**List with contents**

$$\mathsf{listC}(x, \alpha) \;\equiv\; \big(x = \mathsf{nil} \wedge \alpha = \varepsilon \wedge \mathsf{emp}\big) \vee \big(\exists v, z, \alpha'.\, \alpha = v{\cdot}\alpha' \wedge x \mapsto v, z * \mathsf{listC}(z, \alpha')\big)$$




$\varepsilon$: the empty sequence. $v \cdot \alpha'$: $v$ in front of $\alpha'$.

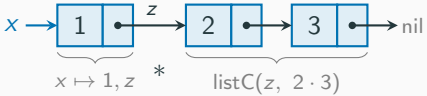
# listC on an example




This heap satisfies listC($x$, $1 \cdot 2 \cdot 3$).

What do we get if we unfold it once?

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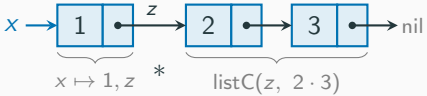



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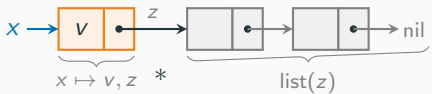
Forget the contents and you get list back:

$$\mathsf{list}(x)\ \equiv\ \exists \alpha.\ \mathsf{listC}(x, \alpha)$$

# Unfolding is the engine in proofs

A predicate is used by unfolding its definition — one step at a time. If we know $x \neq \mathsf{nil}$, then $\mathsf{list}(x)$ unfolds to

$$\exists v, z.\ x \mapsto v, z \ * \ \mathsf{list}(z)$$

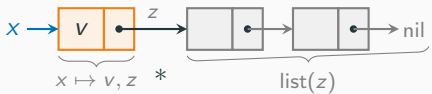



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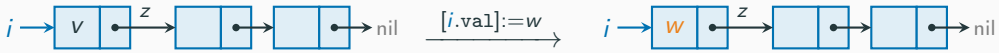



One concrete head cell, exposed; the rest is a shorter list, still folded.

Unfold at the front, operate on the exposed cell, fold back up.

Every list proof is this rhythm. The $*$ ensures the exposed cell and the tail never alias — so we can mutate the head and frame the tail.

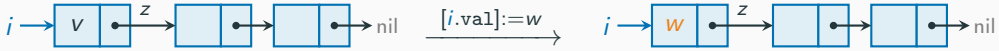
# Example: overwrite the head




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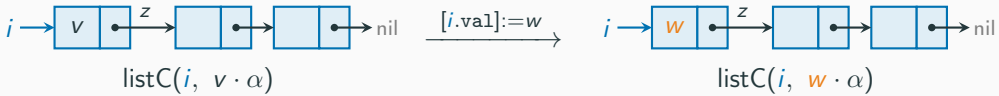

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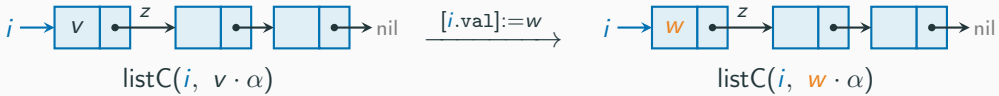
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$\{i \mapsto v, z * \mathsf{listC}(z, \alpha)\}$ unfold (sequence is non-empty)

$[i.\mathtt{val}] := w$ mutation, frame $\mathsf{listC}(z, \alpha)$

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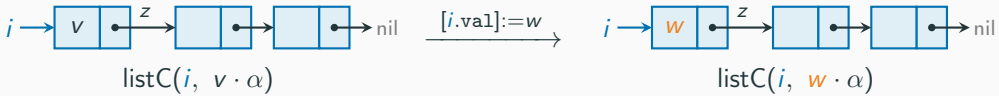
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# In-place list reversal

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list($i$) list($j$)

{list($i$)}

```
j := nil;
while i ≠ nil do {
  k := [i.next];      // remember the rest
  [i.next] := j;      // flip this link
  j := i;             // grow reversed part
  i := k              // advance
}
```

{list($j$)}

- $i$ walks the remaining list; $j$ grows the reversed part.

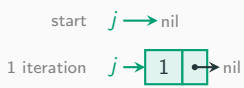
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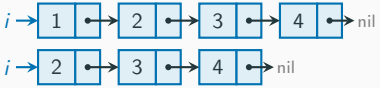
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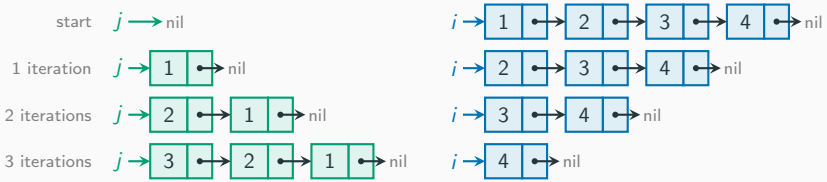
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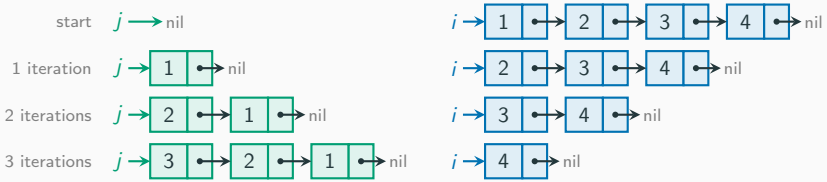
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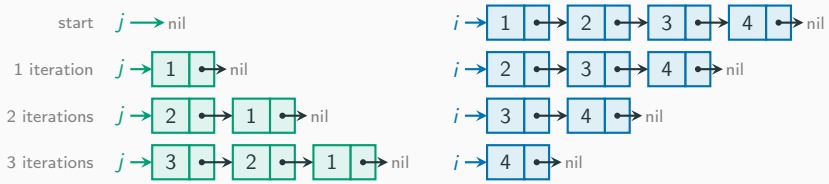
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What is true after every iteration?

# The invariant

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What is true after every iteration?

**Loop invariant**

$$I \equiv \text{list}(i) * \text{list}(j)$$

Two disjoint lists. $i$: what is left to do. $j$: what is done, reversed.

# One iteration, pictured



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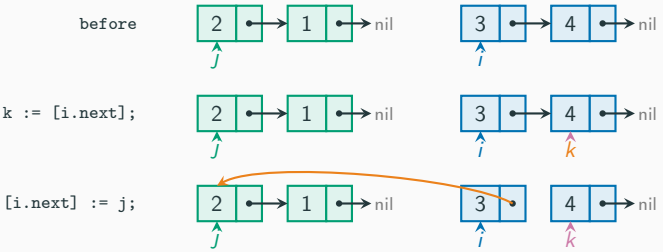
before



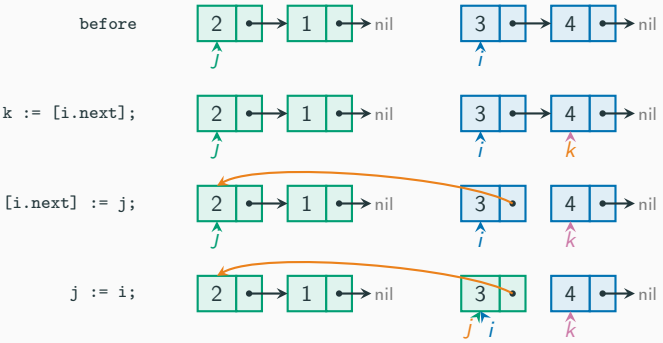

```
k := [i.next];
```



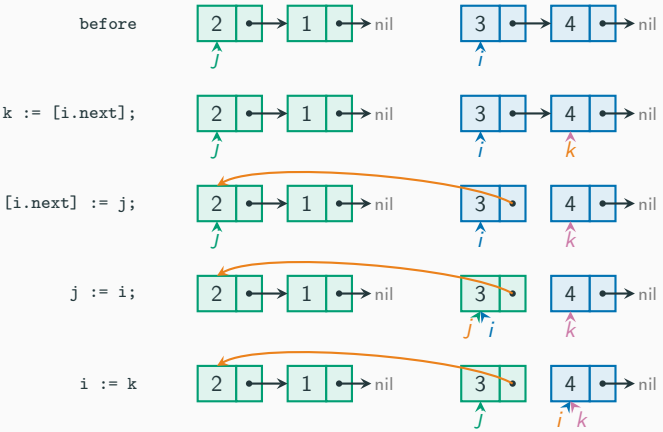
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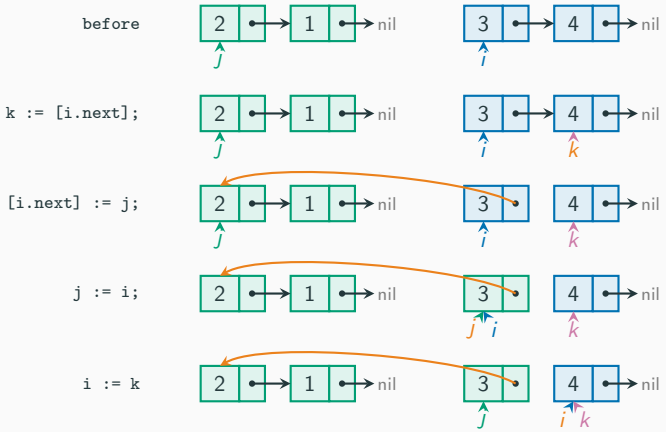
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One link flips; every other cell is untouched. Invariant again: two disjoint lists ✓

# The proof, whole program

Write $I$ for the invariant $\mathsf{list}(i) * \mathsf{list}(j)$.

$\{\mathsf{list}(i)\}$
  $j := \mathsf{nil};$
$\{\mathsf{list}(i) \wedge j = \mathsf{nil}\}$
$\{I\}$ VC1
  while $i \neq \mathsf{nil}$ do
    $\{I \wedge i \neq \mathsf{nil}\}$ *body* $\{I\}$ VC2
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**Verification conditions**

VC1 init: $\mathrm{list}(i) \wedge j = \mathrm{nil} \Rightarrow I$

VC2 preserve: $\{I \wedge i \neq \mathrm{nil}\}$ *body* $\{I\}$

VC3 exit: $I \wedge i = \mathrm{nil} \Rightarrow \mathrm{list}(j)$

# Proving VC1 and VC3

$I \equiv \mathsf{list}(i) * \mathsf{list}(j)$, and $\mathsf{list}(\mathsf{nil}) \equiv \mathsf{lseg}(\mathsf{nil}, \mathsf{nil}) \equiv \mathsf{emp}$.

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VC3: $I \wedge i = \mathsf{nil} \Rightarrow \mathsf{list}(j)$

$$
\begin{array}{lll}
& \mathsf{list}(i) * \mathsf{list}(j) \wedge i = \mathsf{nil} & \\
\Rightarrow & \mathsf{emp} * \mathsf{list}(j) & \text{unfold: } i = \mathsf{nil}, \text{ so } \mathsf{list}(i) \text{ is emp} \\
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# Proving VC2: one iteration

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| $[i.\mathtt{next}] := j;$ | mutation, frame $\mathsf{list}(k) * \mathsf{list}(j)$ |

# Proving VC2: one iteration

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| $\{\mathsf{list}(i) * \mathsf{list}(j) \wedge i \neq \mathsf{nil}\}$ | $I \wedge i \neq \mathsf{nil}$ |
| $\{i \mapsto v, z * \mathsf{list}(z) * \mathsf{list}(j)\}$ | unfold, as $i \neq \mathsf{nil}$ |
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| $\{\mathsf{list}(k) * \mathsf{list}(i)\}$ | fold: $i \mapsto v, j * \mathsf{list}(j)$ |

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| $\{\mathsf{list}(i) * \mathsf{list}(j)\}$ | $= I$ ✓ |

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| $j := i; \quad i := k$ | assignment, twice |
| $\{\mathsf{list}(i) * \mathsf{list}(j)\}$ | $= I$ ✓ |

Side conditions ✓: the lookup modifies only $k$; the mutation modifies nothing.

# Where the frame rule earned its keep

The mutation $[i.\texttt{next}] := j$ used the small axiom on a single cell:

$$\{i \mapsto v, _\}\ \ [i.\texttt{next}] := j\ \ \{i \mapsto v, j\}$$

# Where the frame rule earned its keep

The mutation $[i.\texttt{next}] := j$ used the small axiom on a single cell:

$$\{i \mapsto v, _\}\ \ [i.\texttt{next}] := j\ \ \{i \mapsto v, j\}$$

Everything else in the heap was framed out:

$$\{(i \mapsto v, _) * \underbrace{\mathsf{list}(k) * \mathsf{list}(j)}_{\text{framed}}\}\ [i.\texttt{next}] := j\ \{(i \mapsto v, j) * \mathsf{list}(k) * \mathsf{list}(j)\}$$

# Is this full functional correctness?

We proved

$$\{\mathsf{list}(i)\} \quad \mathsf{reverse} \quad \{\mathsf{list}(j)\}$$

Does this say that reverse reverses the list?

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j := i;  i := nil
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No. This program meets the same spec:

```
j := i;  i := nil
```

How would you fix the spec?

$$\{\mathsf{listC}(i, \alpha_0)\} \quad \mathsf{reverse} \quad \{\mathsf{listC}(j, \ \mathrm{rev}(\alpha_0))\}$$

# Separation logic at scale: Infer

# Infer

A static analyser from Meta, open source since 2015 (`fbinfer.com`).

- Built on separation logic. Began as a startup (Monoidics) by Calcagno, Distefano, O'Hearn; acquired by Facebook in 2013.
- Runs on code changes at Meta.
- Java, C, C++, Objective-C. Finds null dereferences, memory and resource leaks.



The Monoidics team at Facebook. Photo: QMUL Theory group.

# Infer on a list program

```
#include <stdlib.h>
struct node { int val; struct node *next; };

int second(struct node *x) {
  return x->next->val;
}

int caller(void) {
  struct node n = {1, NULL};
  return second(&n);
}

struct node *push(struct node *l, int v) {
  struct node *n = malloc(sizeof *n);
  if (n == NULL) return l;
  n->val = v;  n->next = l;
  return n;
}

void drop(void) {
  push(NULL, 1);
}
```

- No annotations: no pre/post, no invariants.
- Infer infers a spec for each function, in separation logic.
- At a call, the callee's spec is reused; the rest of the heap is framed.

Infer reports two bugs. Where are they?

# Infer on a list program

```
#include <stdlib.h>
struct node { int val; struct node *next; };

int second(struct node *x) {
  return x->next->val;
}

int caller(void) {
  struct node n = {1, NULL};
  return second(&n);            // null pointer dereference
}

struct node *push(struct node *l, int v) {
  struct node *n = malloc(sizeof *n);
  if (n == NULL) return l;
  n->val = v;  n->next = l;
  return n;
}

void drop(void) {
  push(NULL, 1);                // memory leak
}
```

- No annotations: no pre/post, no invariants.
- Infer infers a spec for each function, in separation logic.
- At a call, the callee's spec is reused; the rest of the heap is framed.

# Running it

```
$ infer run -- gcc -c list.c
```

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```
$ infer run -- gcc -c list.c



list.c:10: error: Null Dereference
  `n.next` could be null (null value originating from line 9) and is
  dereferenced in the call to `second()`.
  10.   return second(&n);            // null pointer dereference
                ^

list.c:21: error: Memory Leak
  Memory dynamically allocated by `malloc`, indirectly via call to `push()`
  on line 21 is not freed after the last access at line 21, column 3.
  21.   push(NULL, 1);                // memory leak
        ^

Found 2 issues
```

# Wrap

# Today

- Inductive predicates — $\mathsf{lseg}(x, y)$, $\mathsf{list}(x)$ — describe unbounded heap structures.
  $*$ in the definition bakes in well-formedness.
- Proof rhythm: unfold, operate, fold.
- In-place list reversal.
- Infer: separation logic, at Meta scale, with no annotations.

# Next time

We have been proving programs correct — with invariants we supply.

Lecture 12: instead of *proving correctness* we try to find bugs.

# Questions?

ATiPL 2026 Minute Sheet